

Inventory Control with Stochastic and Non-Stationary Demand

- La domanda nell'unità di tempo è una variabile aleatoria x con valore atteso $E(x)$ e deviazione standard $\sigma(x)$, che cambiano nel tempo (domanda stocastica e non stazionaria in media e varianza).
- Siccome $E(x)$ e $\sigma(x)$ sono tempo-dipendenti, occorrono dei previsori per entrambi.

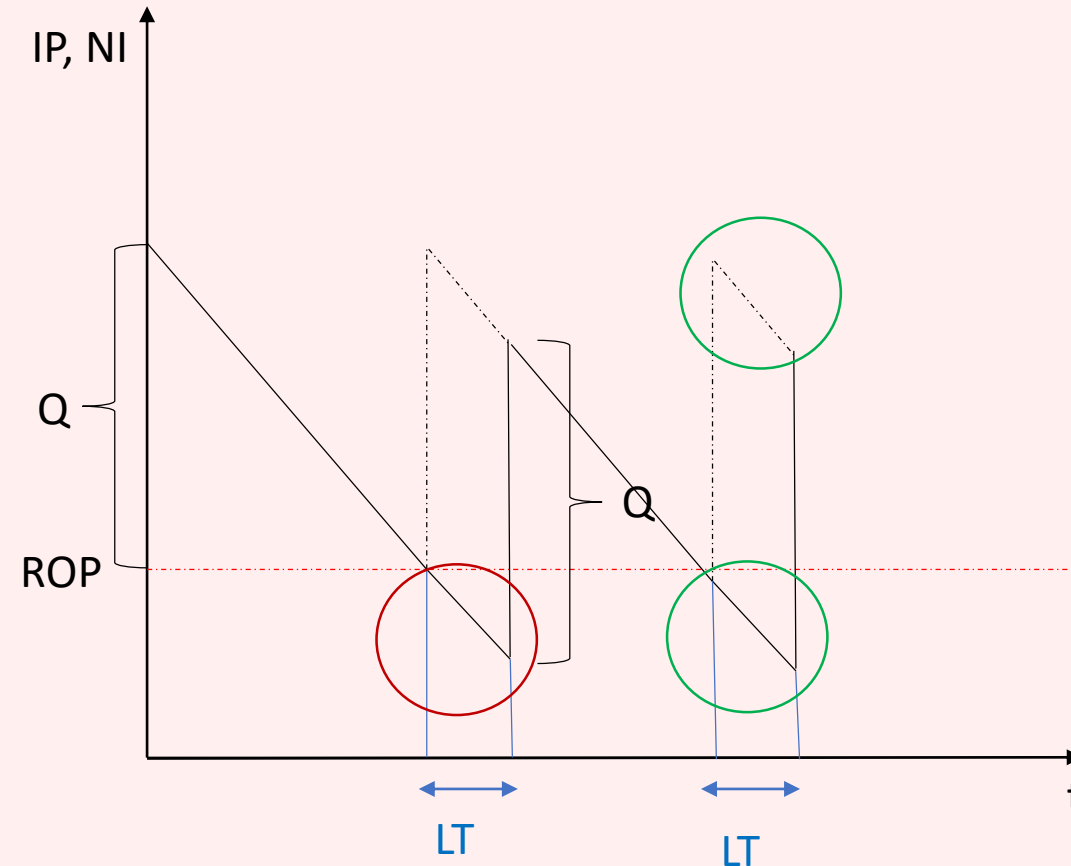
Inventory Control with Stochastic and Non-Stationary Demand: (ROP,Q)

Inventory Position = IP
Giacenza virtuale che include anche i
backorder (può essere negativa)

Net Inventory = NI
Giacenza fisica

NOTA: Gli ordini si lanciano in base a IP, **NON** NI

ORDINO SE: $IP < ROP$



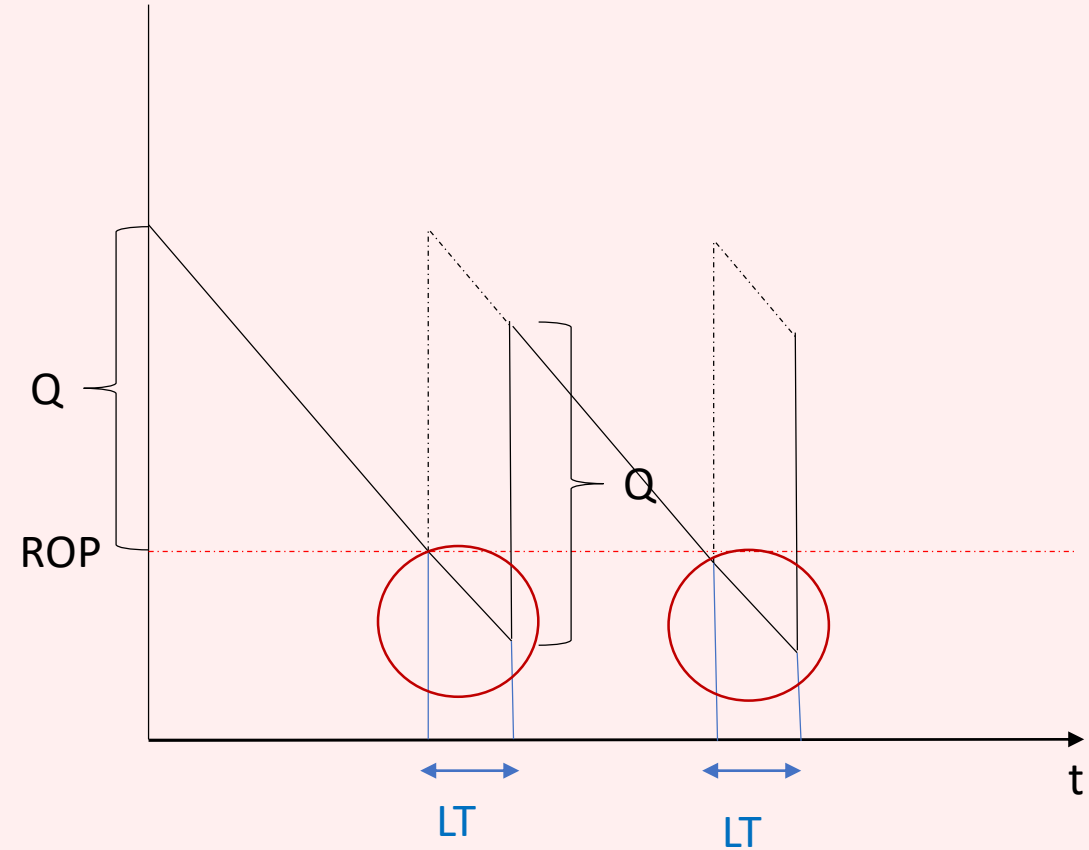
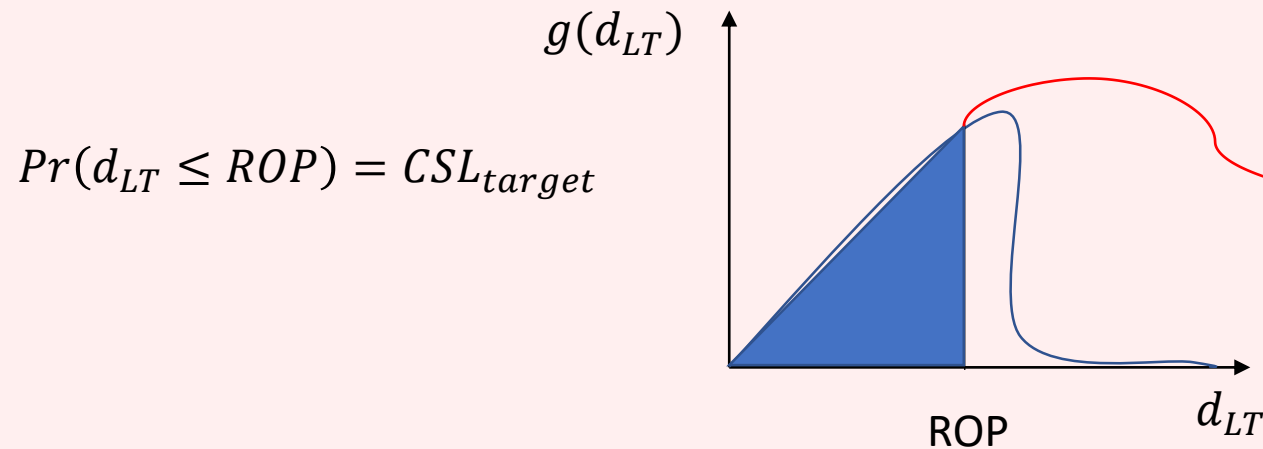
Inventory Control with Stochastic and Non-Stationary Demand: (ROP,Q)

Come trovo ROP e Q?



Impongo un **Cycle Service Level** = CSL target

Probabilità di andare sotto scorta durante LT



$$G(ROP) = CSL_{target}$$



$$ROP = G^{-1}(CSL_{target})$$

Inventory Control with Stochastic and Non-Stationary Demand: (ROP,Q)

HP: $g(d_{LT}) \sim N(\mu_{LT}, \sigma_{LT})$

$$ROP = \mu_{LT} + SS$$

$$ROP = \mu_{LT} + k * \sigma_{LT}$$

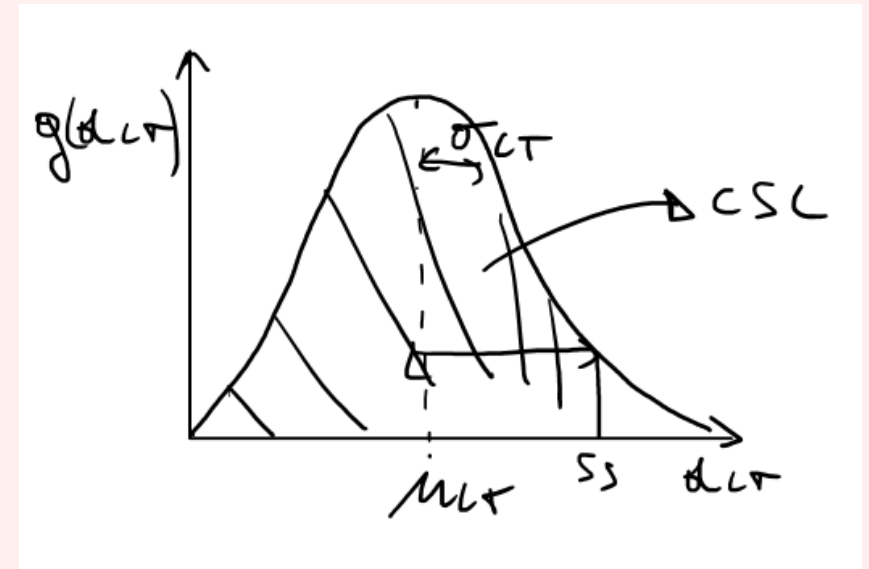
K = Safety factor (valori tabulati)

È ricavato, noto il CSL, a partire dalla distribuzione normale standardizzata $N(0,1)$.

$$CSL = 68,26\% \quad k = 1$$

$$CSL = 95,44\% \quad k = 2$$

$$CSL = 99,73\% \quad k = 3$$



$$ROP = \mu_{LT} + \Phi^{-1}(CSL_{target}) * \sigma_{LT}$$

$$k = \Phi^{-1}(CSL_{target}) = \text{INV. NORM. S}(CSL_{target})$$

Inventory Control with Stochastic and Non-Stationary Demand: (ROP,Q)

(ROP,Q) DINAMICO $\Rightarrow$ (ROP_t, Q_t)

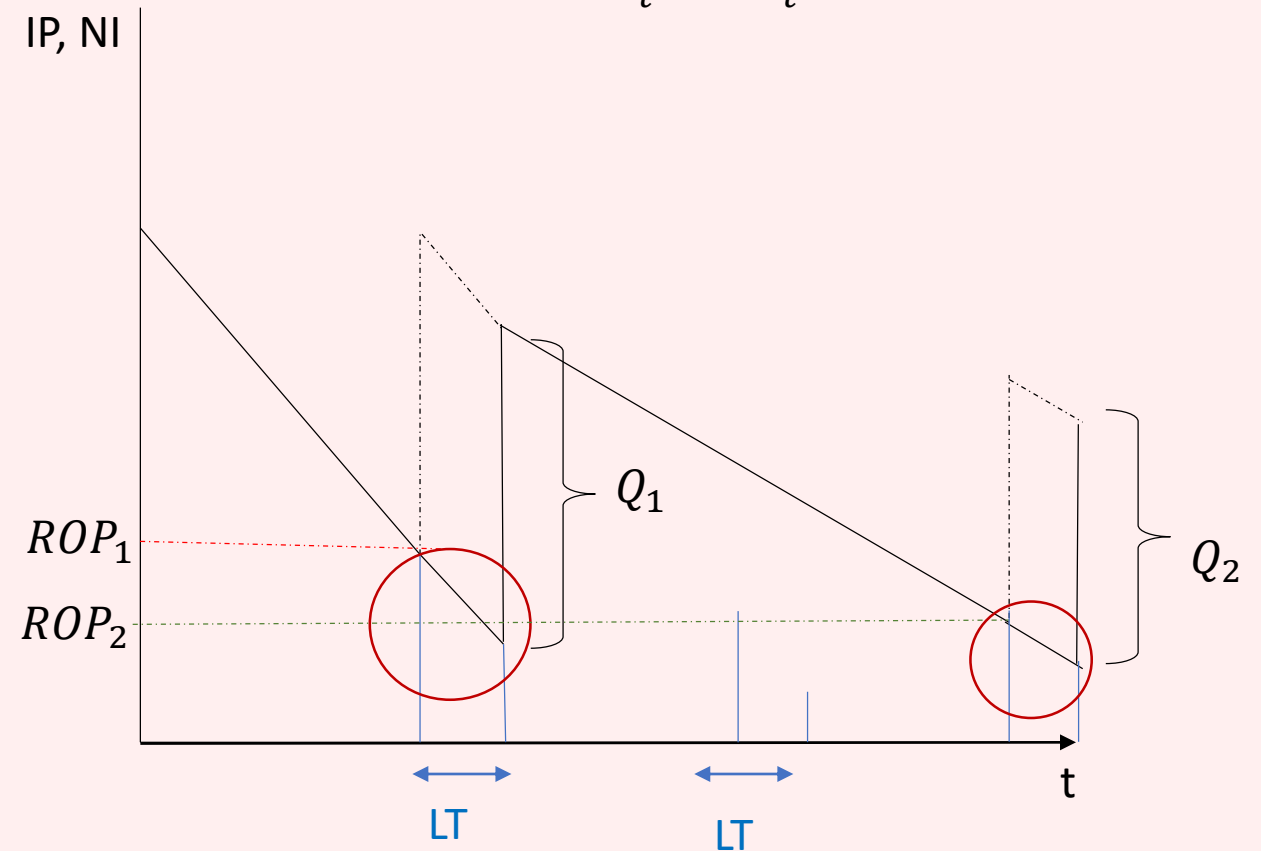
$$ROP_t = \mu_{LT_t} + \Phi^{-1}(CSL_{target}) * \sigma_{LT_t}$$

$\Rightarrow$ Dinamicità del sistema $\Rightarrow$ In ogni periodo ho i previsori $\hat{\mu}_{LT_t}$ e $\hat{\sigma}_{LT_t}$

Come stimare μ_{LT_t} e σ_{LT_t} ?



Previsori $\hat{\mu}_{LT_t}$ e $\hat{\sigma}_{LT_t}$



Inventory Control with Stochastic and Non-Stationary Demand: (ROP,Q)

$\hat{\sigma}_{LT} ?$



$$\sigma_{LT} = \sqrt{VAR * LT}$$

Varianza della domanda a periodo

Numero di periodi di cui è composto LT



$$\hat{\sigma}_{LT} = \sqrt{\widehat{MSE} * LT}$$

con $\widehat{MSE}$ stimatore della varianza in un periodo



Mi copro per tutta la domanda attesa
che ho previsto+1,2,3.. volte l'errore
che prevedo di fare nella previsione

$\widehat{MSE} ?$



Smorzamento esponenziale semplice

$$\widehat{MSE}_t = \gamma * (d_t - D_t)^2 + (1 - \gamma) * \widehat{MSE}_{t-1}$$

Inventory Control with Stochastic and Non-Stationary Demand: (T,S)

----- Inventory Position =IP

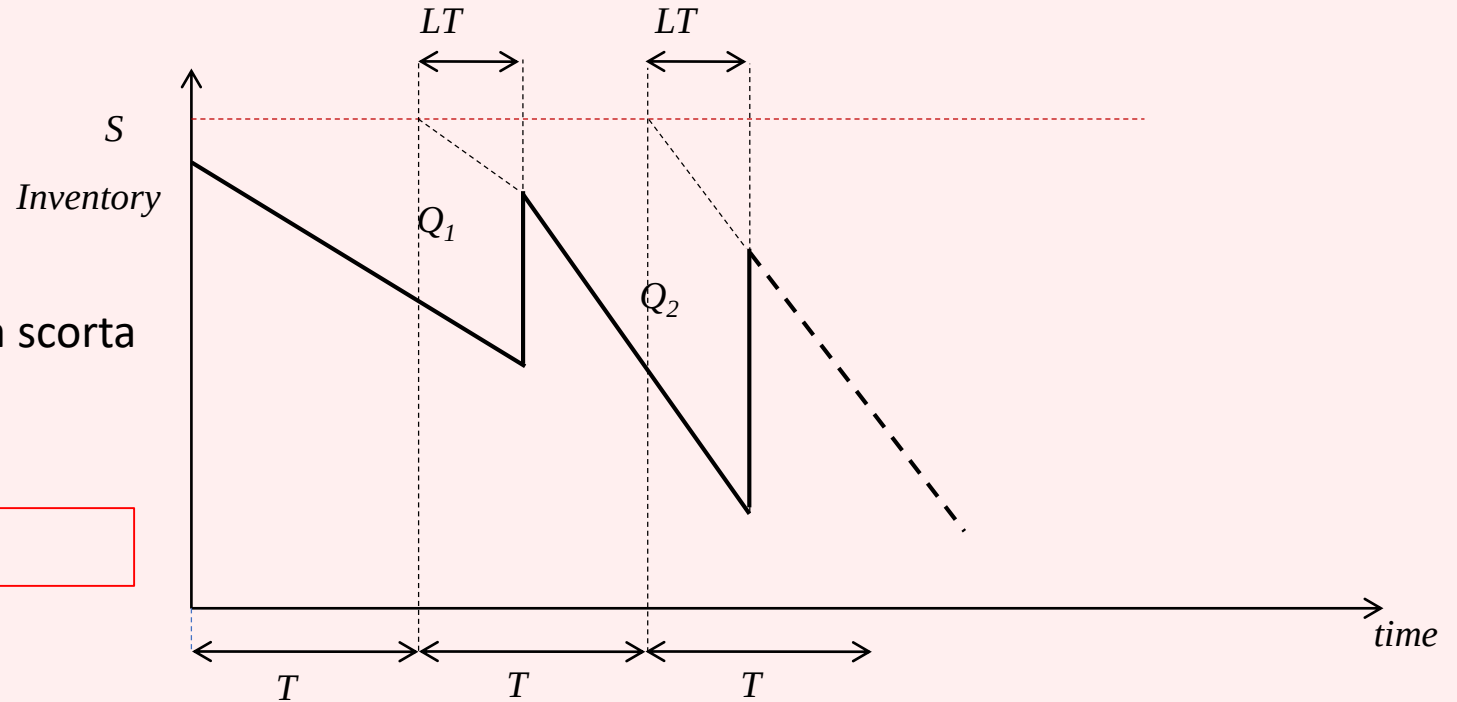
———— Net Inventory =NI

Si lancia l'ordine ogni T periodi per riportare la scorta al livello S (Order-Up-To Level)

S deve garantire CSL_{target} in $T+LT$

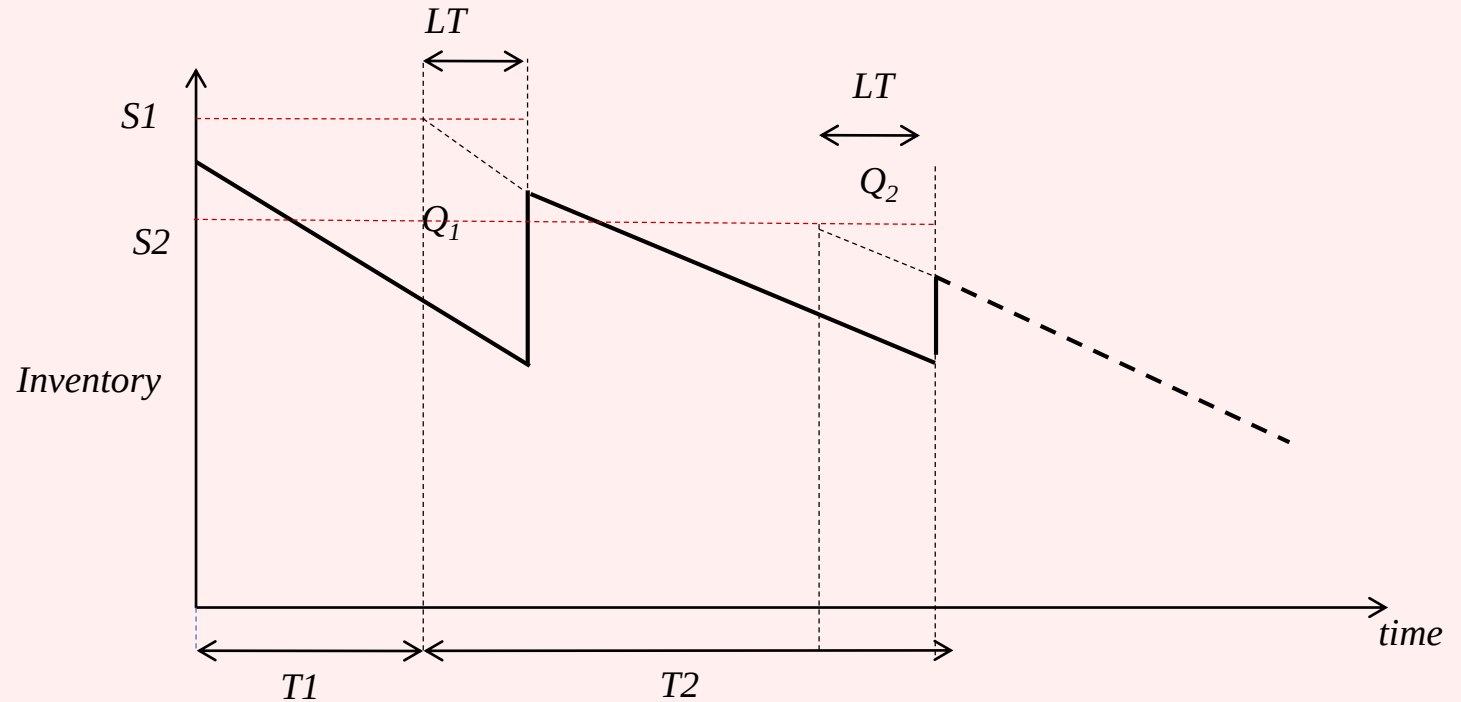
Infatti si ordina in T ma la merce arriva in $T+LT$

$$S = G^{-1}_{T+LT} (CSL_{target})$$



Inventory Control with Stochastic and Non-Stationary Demand: (T,S)

(T,S) DINAMICO $\rightarrow$ (T, S_t)



$$HP: g(d_{T+LT}) \sim N(\mu_{T+LT}, \sigma_{T+LT})$$

$$S = \mu_{T+LT} + SS \quad \rightarrow \quad S = \mu_{T+LT} + SS \quad \rightarrow \quad S = \mu_{T+LT} + \Phi^{-1}(CSL_{target}) * \sigma_{T+LT}$$

Inventory Control with Stochastic and Non-Stationary Demand: (T,S)

$$S = \mu_{T+LT} + \Phi^{-1}(CSL_{target}) * \sigma_{T+LT} \quad \text{Come stimare } \mu_{T+LT_t} \text{ e } \sigma_{T+LT_t}?$$



Previsori $\widehat{\mu_{T+LT_t}}$ e $\widehat{\sigma_{T+LT_t}}$

$$\widehat{\mu_{T+LT_t}} = \widehat{D}_t * (LT + T)$$

$$\widehat{\sigma_{T+LT_t}} = \sqrt{\widehat{MSE}_t * (LT + T)}$$

con:

$\widehat{D}_t$ previsione della domanda a periodo

$\widehat{MSE}_t = \gamma * (d_t - D_t)^2 + (1 - \gamma) * \widehat{MSE}_{t-1}$ previsione della varianza della domanda a periodo